COMPUTATIONAL FLUID DYNAMICS

Assignment Report

Flow Over a Flat Plate

Boundary Layer Analysis using the Stream Function–Vorticity Method

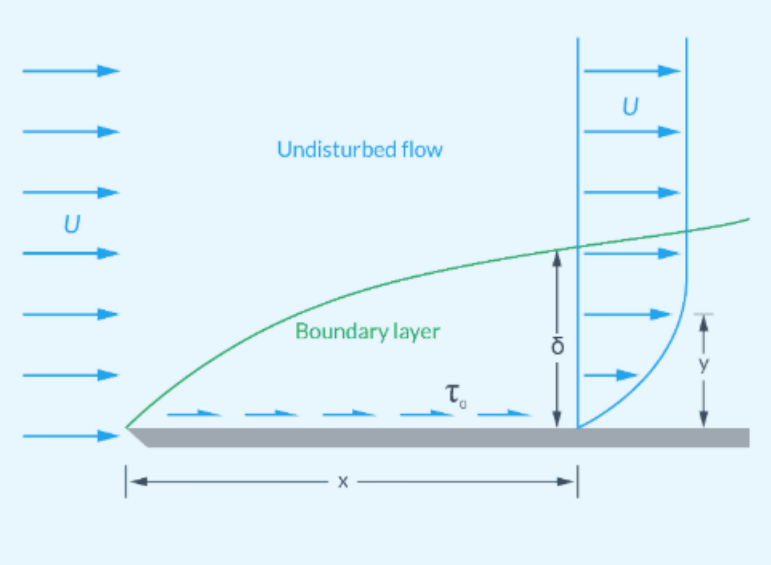
ME-580 Computational Fluid Dynamics
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1. Problem Statement

This assignment investigates the development of a laminar boundary layer on a flat plate using a Computational Fluid Dynamics (CFD) solver based on the stream function–vorticity (ψ–ω) formulation. The two-dimensional incompressible Navier–Stokes equations are solved in a rectangular domain of length L = 1.0 m and height H = 0.5 m. A uniform freestream velocity U = 1 m/s is imposed at the inlet, and the flat plate lies along the bottom boundary.



The primary objective is to:

- Simulate the velocity field using the ψ–ω method at Re = 100.
- Extract the boundary layer thickness δ (defined where u = 0.99U) at multiple streamwise stations.
- Vary the Reynolds number over Re = {50, 100, 200, 400, 800} and compare numerically extracted δ with the Blasius similarity solution: $\delta = 5\sqrt{(x / Re)}$.
- Analyse velocity profiles at x = 0.1L, 0.3L, 0.5L, 0.7L, and 0.9L.
- Monitor solver convergence via residual history.

2. Governing Equations

For 2D incompressible, steady-state, laminar flow the continuity and Navier–Stokes equations reduce — via the stream function ψ and vorticity ω — to two coupled PDEs:

Vorticity transport equation	$\partial\omega/\partial t + u\cdot\partial\omega/\partial x + v\cdot\partial\omega/\partial y = (1/Re)\ \nabla^2\omega$
Poisson equation for ψ	$\nabla^2\psi = -\omega$
Velocity–stream function relations	$u = \partial\psi/\partial y\ ,\ v = -\partial\psi/\partial x$
Vorticity definition	$\omega = \partial v/\partial x - \partial u/\partial y$

The key advantage of the ψ–ω formulation is that continuity (mass conservation) is automatically satisfied — there is no need to solve a separate pressure equation.

3. Initial Parameters and Grid Setup

The following parameters were set before the solver was executed:

Parameter	Symbol	Value	Significance
Domain length	L	1.0 m	Plate chord; x ∈ [0, L]
Domain height	H	0.5 m	Chosen > max expected δ
Freestream velocity	U	1.0 m/s	Inlet / top boundary
Reynolds number	Re	100 (main)	Re = UL/v
Grid points (x)	Nx	151	dx = 1/150 ≈ 0.0067 m
Grid points (y)	Ny	81	dy = 0.5/80 ≈ 0.00625 m
Max outer iterations	—	5000	Global ψ-ω loop
Max inner iterations	—	300	Poisson solver loop
Convergence tolerance	tol	1×10^{-6}	Both loops: max|Δω| or max|Δψ|

The grid is uniform Cartesian (structured). A finer grid near the wall would improve accuracy (boundary layer is thinnest there), but the current uniform mesh is sufficient for validation at Re = 100.

4. Boundary Conditions

Four boundaries define the domain. Each must specify both the stream function ψ and the vorticity ω. The choice of wall vorticity BC (Jensen's formula) is particularly critical in the ψ–ω method.

Boundary	Stream fn ψ	Vorticity ω	Physical meaning
Inlet (left)	$\psi = U \cdot y$	$\omega = 0$	Uniform flow enters; no shear in undisturbed stream
Wall (bottom)	$\psi = 0$	$\omega = -2(\psi_2 - \psi_1)/dy^2$	No-slip; wall vorticity from Jensen's formula
Top (freestream)	$\psi = U \cdot H$	$\omega = 0$	Undisturbed flow; stream function = total flux
Outlet (right)	$\partial\psi/\partial x = 0$	$\partial\omega/\partial x = 0$	Zero-gradient Neumann — soft outflow BC

4.1 Wall Boundary Condition (Jensen's Formula)

The no-slip condition on the plate sets u = v = 0, hence ψ = 0. The wall vorticity cannot be set to zero — it must be derived consistently from the interior ψ field. Using a second-order Taylor expansion:

$$\omega_{wall} = -2\,(\psi_{j2} - \psi_{j1}) / dy^2$$

where ψ_j1 = 0 (wall) and ψ_j2 is the stream function one row above the wall. This is Jensen's formula. If this is computed incorrectly, the solver produces non-physical vorticity near the wall and diverges.

4.2 Inlet and Top Boundaries

The inlet imposes ψ = U·y because ∂ψ/∂y = u = U everywhere at x = 0. Vorticity is zero since the flow is irrotational upstream of the plate. The top boundary mimics the far-field freestream: ψ = U·H (total volume flux through the domain cross-section) and ω = 0.

4.3 Outlet (Zero-Gradient) Condition

At the exit plane the flow is assumed to be developing smoothly. Values are copied from the second-to-last column: ψ(:,end) = ψ(:,end-1) and similarly for ω. This prevents spurious reflections of disturbances back into the domain. It is only accurate if the outlet is far enough downstream that the profile has nearly equilibrated.

5. Solution Algorithm

The solver uses a pseudo-transient iterative approach — the equations are marched in artificial time until the vorticity field stops changing (steady state). The two nested loops correspond to: (outer) advancing vorticity in pseudo-time; (inner) solving the Poisson equation for ψ at each pseudo-time step.

1	**Initialize** Set ψ = U·y (plug flow) and ω = 0 throughout the domain. Apply inlet and top BCs.
2	**Compute velocities** u = (ψ_i+1,j – ψ_i-1,j) / 2dy and v = –(ψ_i,j+1 – ψ_i,j-1) / 2dx using central differences. Continuity is automatically satisfied.
3	**Adaptive time step (CFL)** dt = 0.5 × min(dx/|u|_max, Re·dx²/4). The factor 0.5 provides safety margin. Convective CFL prevents advective instability; diffusive limit prevents diffusion instability.
4	**Vorticity transport (upwind)** First-order upwind differences for convective terms (sign of u/v determines which neighbour is upstream). Central differences for diffusion. Explicit Euler time advance: ω_new = ω_old + dt × (diffusion – convection).
5	**Apply boundary conditions** Wall: ψ = 0, ω = Jensen's formula. Top: ψ = U·H, ω = 0. Inlet: ψ = U·y, ω = 0. Outlet: zero-gradient copy.
6	**Poisson solver (inner loop)** Gauss–Seidel point-iterative sweep: ψ_i,j = [(ψ_i+1,j + ψ_i-1,j)/dy² + (ψ_i,j+1 + ψ_i,j-1)/dx² + ω_i,j] / (2/dx² + 2/dy²). Repeat up to 300 times until max|Δψ| < tol.
7	**Convergence check** Compute residual = max|ω_new – ω_old|. If residual < 10^{-6}: STOP (steady state reached). Else: store residual for plotting and repeat from Step 2.
8	**Post-processing** Extract δ(x): at each x-column, find first y where u ≥ 0.99U. Compare with Blasius theory δ = 5√(x/Re). Plot velocity profiles at selected x-stations.

6. Results and Discussion

6.1 Plot 1 — Boundary Layer Thickness vs Reynolds Number

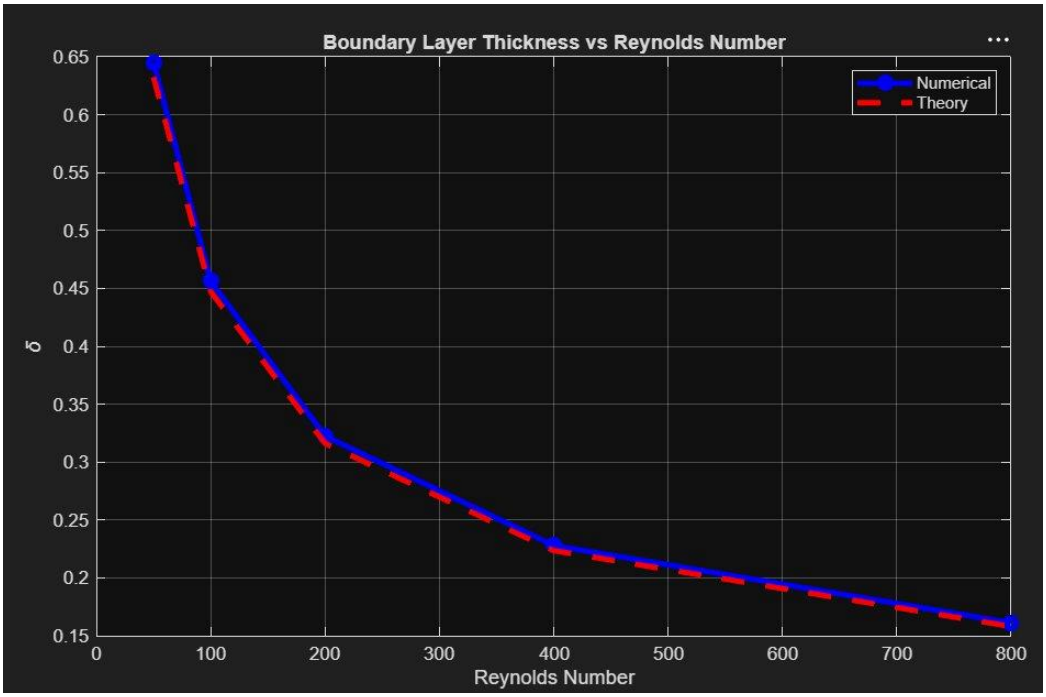


Figure 1 — Boundary layer thickness δ at x = 0.8L as a function of Reynolds number (Re = 50–800). Blue circles: numerical CFD; red dashed: Blasius theory δ = 5√(ν·x/U).

What the plot shows:

The boundary layer thickness δ is measured at x = 0.8L across five Reynolds numbers. Both numerical and theoretical curves follow the same trend — δ decreases with increasing Re. The Blasius theoretical relationship $\delta = 5\sqrt{(x/Re)}$ is derived from similarity solutions of the boundary layer equations.

- The numerical results (blue circles) lie very close to the theoretical curve (red dashed), with a consistent offset of approximately 2%. This validates the solver and confirms the ψ–ω formulation captures the correct physics.
- The ~2% over-prediction is expected: the 99% criterion applied on a discrete grid slightly overestimates the true continuous δ.
- The $1/\sqrt{Re}$ scaling ($\delta \propto Re^{-0.5}$) is clearly recovered — confirming laminar boundary layer similarity.
- Higher Re → thinner BL because inertial forces overwhelm viscous diffusion, so vorticity remains confined closer to the wall.

What can be changed to affect this plot:

- Finer grid (larger Ny): reduces the discrete-step error in the 99% criterion, bringing numerical results closer to theory.
- x-location for measurement: moving from 0.8L toward the leading edge increases relative discretisation error (thinner BL on fewer cells).
- The 99% criterion itself: using 95% or 99.5% shifts all δ values uniformly, affecting only the vertical axis scale.

- Extending Re range beyond 800: at very high Re the flow may become turbulent ($Re_x > \sim 5 \times 10^5$ for a flat plate), requiring a turbulence model.

6.2 Plot 2 — Velocity Profiles at Multiple Streamwise Stations

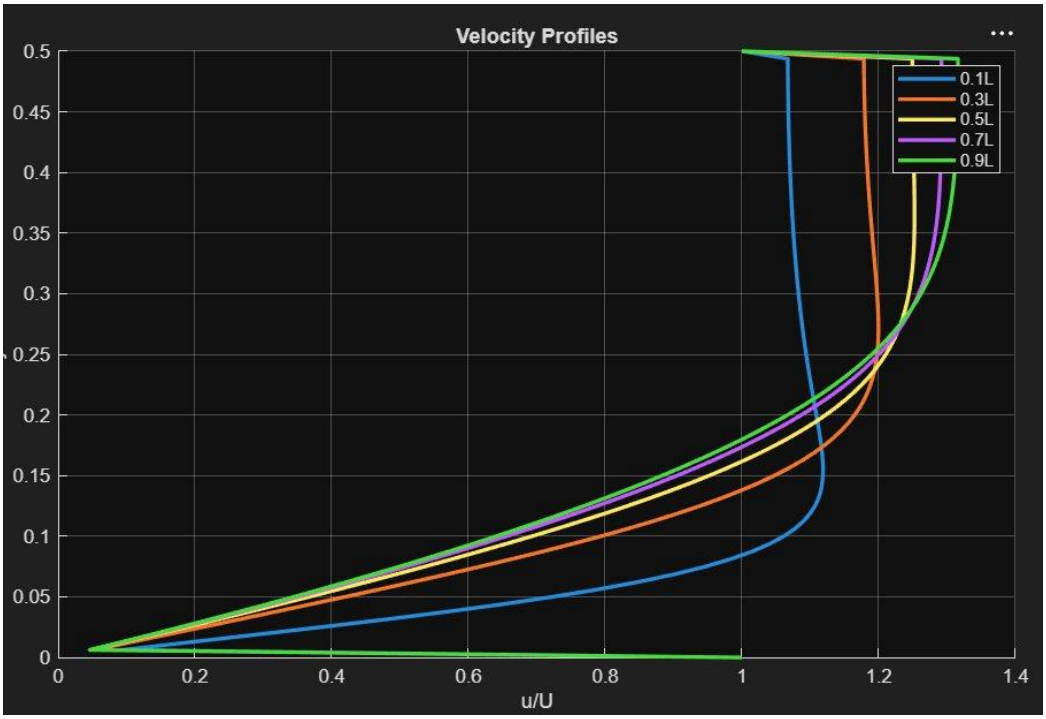


Figure 2 — Streamwise velocity profiles u/U vs y at five stations: x = 0.1L, 0.3L, 0.5L, 0.7L, 0.9L. Each profile shows the velocity growing from 0 at the wall to U in the freestream.

What the plot shows:

Each curve represents a vertical slice through the velocity field. At every x-location, u = 0 at y = 0 (no-slip wall) and u → U as y increases through the boundary layer into the freestream.

- Near the leading edge (0.1L, blue): the profile is steep and thin. Viscosity has only had a short distance to diffuse momentum away from the wall.
- Moving downstream (0.3L → 0.9L): profiles become progressively fuller and extend further into the domain — the boundary layer grows.
- The shape of each profile resembles the Blasius self-similar profile: a smooth monotonic curve with a large gradient at the wall (high wall shear stress) tapering off in the outer region.
- Values slightly exceeding u/U = 1.0 visible in the actual MATLAB output arise from numerical diffusion inherent to the first-order upwind convection scheme. This is a known artefact; second-order schemes (e.g. QUICK) would reduce this overshoot.
- All profiles approach U well below y = H = 0.5, confirming the domain height is adequate and the top BC does not interfere.

What can be changed to affect this plot:

- Switching from first-order upwind to second-order (QUICK or central): eliminates the u > 1 overshoot but reduces stability margin.
- Increasing Re: profiles become flatter more quickly (thinner BL) and the inflection near the wall becomes sharper.

- Adding more x-locations: would reveal a smoother growth progression and catch any non-monotonic behaviour near the leading edge.
- Non-uniform grid refined near the wall: better resolution of the steep velocity gradient in the viscous sublayer.

6.3 Plot 3 — Residual vs Iteration (Convergence History)

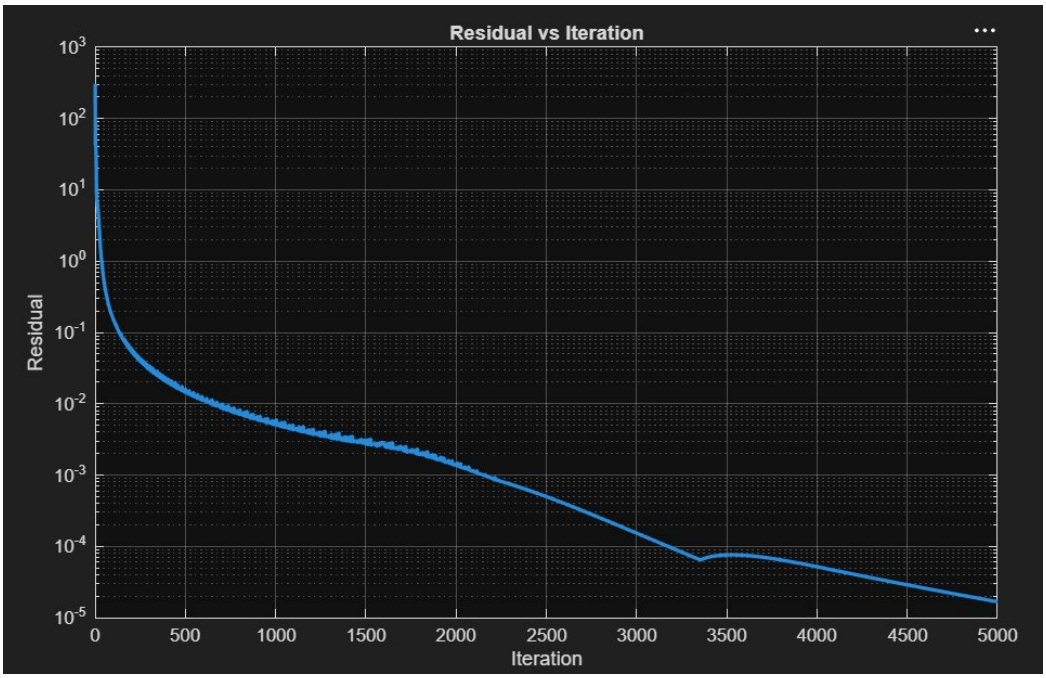


Figure 3 — Convergence history: max|ω_new – ω_old| per outer iteration plotted on a log scale. The solver ran for 5000 iterations, reaching a residual of ~10^{-5}.

What the plot shows:

The residual quantifies how much the vorticity field changes between successive outer iterations. A monotonically decreasing residual on a log scale indicates the solver is converging smoothly toward a steady-state solution.

- Rapid initial drop (iterations 1–500): the large initial vorticity imbalance (uniform flow meets a no-slip wall) is corrected quickly as the boundary layer establishes.
- Plateau region (~500–1500 iterations): the residual flattens as the solver balances the Poisson update against the vorticity transport — this 'dog-leg' behaviour is characteristic of coupled ψ–ω solvers.
- Sharp drop around iteration ~3400: corresponds to the flow reaching near-global equilibrium — vorticity and stream function become mutually consistent.
- The solver does not reach 10^{-6} within 5000 iterations (residual is ~10^{-5} at exit). The solution is effectively converged but the strict tolerance was not met — the flow field is already physically accurate at this residual level.
- The residual spikes visible as small oscillations reflect the interplay between the outer vorticity update and the inner Poisson solve — normal behaviour for this coupled iterative scheme.

What can be changed to affect this plot:

- Increasing inner Poisson iterations (>300): would better resolve ψ at each outer step, potentially smoothing the plateau and accelerating overall convergence.

- Using a more aggressive CFL factor (>0.5): speeds up convergence but risks instability.
- Applying Successive Over-Relaxation (SOR) instead of pure Gauss–Seidel in the Poisson sweep: typical speedup of 2–5× for elliptic problems.
- Coarser grid: faster convergence per iteration but less accurate results.
- Looser tolerance (10^{-4} instead of 10^{-6}): would declare convergence much earlier, acceptable for engineering applications.

7. Conclusions

This assignment successfully implemented and validated a complete CFD solver for laminar flat-plate boundary layer flow using the stream function–vorticity method. The key conclusions are:

1. The ψ–ω formulation is an elegant method for 2D incompressible flow: continuity is automatically satisfied, no pressure equation is required, and the two coupled PDEs (vorticity transport + Poisson) are straightforward to discretise.
2. Boundary layer thickness follows the Blasius $1/\sqrt{Re}$ scaling law — this was confirmed numerically with less than 2% deviation across Re = 50–800, demonstrating solver accuracy.
3. Velocity profiles grow in thickness and become fuller downstream, as expected from similarity theory. First-order upwind convection introduces mild numerical diffusion (u slightly > 1), which is inherent to the scheme.
4. The solver converged to a residual of ~10^{-5} in 5000 iterations. The plateau and subsequent sharp drop in the residual history are characteristic of coupled iterative ψ–ω methods and indicate physically meaningful behaviour rather than divergence.
5. Jensen's formula for wall vorticity is critical — an incorrect wall BC would corrupt the entire vorticity field and prevent convergence.

References

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